

Anomaly Detection in Enron's Network

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May 15th, 2025

- This project used a real-world dataset of emails from the Enron Corporation
- Involved in one of the biggest corporate fraud scandals in US history
- There are around 700 000 emails exchanged in the relevant time frame
- The goal was to identify anomalies in emails using deep learning techniques
- The detected anomalies were compared against the known timeline of key events in the scandal

Key Events Leading to the Enron Collapse: Part 1

Date	Event Description
Sep 2000	First doubts about Enron's accounting appear in a Wall Street Journal article.
Mar 2001	Fortune publishes "Is Enron Overpriced?"; California investigates market manipulation.
Jun 2001	Internal tensions rise between CEO Skilling and Chairman Lay.
Aug 2001	Skilling resigns; VP Sharon Watkins warns about accounting irregularities.
Oct 22, 2001	SEC announces a preliminary investigation into Enron's financial practices.
Oct 23, 2001	Auditors begin destroying internal documents related to Enron.

Key Events Leading to the Enron Collapse: Part 2

Date	Event Description
Oct 31, 2001	SEC opens a formal investigation; Enron's stock price drops.
Nov 8–9, 2001	Enron restates earnings and announces a possible acquisition by Dynegy.
Nov 15–19, 2001	Credit rating falls; Enron must repay \$690M by Nov 26.
Nov 28, 2001	Dynegy cancels the deal; Enron's stock falls below \$1.
Dec 2, 2001	Enron files for bankruptcy.
Dec 3–12, 2001	4,000 layoffs; lawsuits begin

Example of a data object

Label	Data
sender	veronica.espinoza@enron.com
receiver	michael.etringer@enron.com
date	Mon, 15 Oct 2001 08:30:53 (PDT)
subject	Credit Watch List-Week of 10/15/01
body	Attached is a revised Credit Watch listing for the week of 10/15/01. Please note that Bethlehem Steel Corporation was placed on "NO TRADES". Progas, Inc. and Sanchez Oil & Gas Corporation were placed on "Call Credit". If there are any personnel in your group that were not included in this distribution, please insure that they receive a copy of this report.
folder	deleted_items
week	2001-10-15 00:00:00

Section 1

First Approach: Graphs

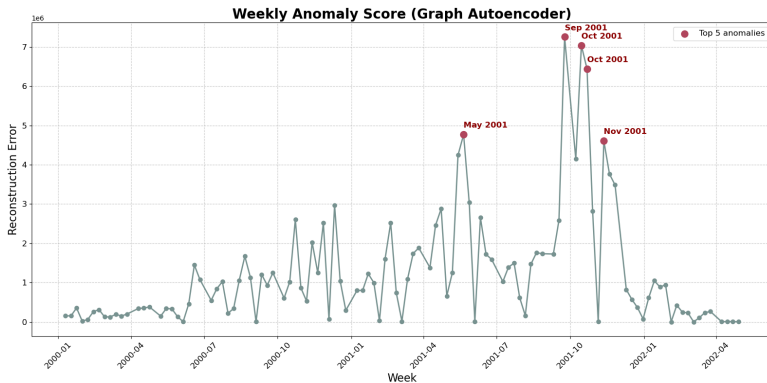
Methods Applied:

- **Graph Autoencoder (GAE)**
- **Structural Metrics Comparison**
- **Autoencoder on Structural Metrics**
- **Node Embedding evolution over time**
- **Dynamical Graph Autoencoder**
- **Ensemble method**

Graph Autoencoder (GAE)

- A GAE is trained to reconstruct the adjacency matrix of a graph.
- It uses Graph Convolutional Networks (GCN) to compress node information into embeddings. (GCNConv)
- The model tries to rebuild the original graph from these embeddings.
- Anomaly score: The higher the reconstruction error, the more anomalous the graph is.

Results: Graph Autoencoder

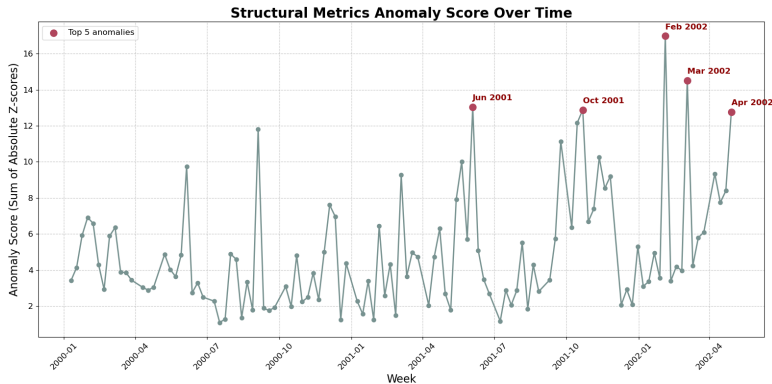


- Major anomaly peaks are observed in:
 - May 2001 (rising internal tensions)
 - September–October 2001 (financial restatements, organizational crisis)
 - November 2001 (SEC investigation)

Structural Metrics Used and Combination Strategy

- Metrics calculated for each weekly graph:
 - **Number of nodes and edges:** Size and activity of the network.
 - **Graph density:** How densely the nodes are connected.
 - **Average clustering coefficient:** Tendency of nodes to form tightly knit groups.
 - **Degree assortativity:** Tendency of nodes to connect with others of similar degree.
 - **Average degree:** Average number of connections per node.
 - **Number of strongly connected components:** Number of groups where every node can reach every other node.
- Combination Strategy:
 - Calculate standardized z-scores for each metric.
 - Define the anomaly score as the sum of absolute z-scores.

Results: Structural Metrics Anomaly Score

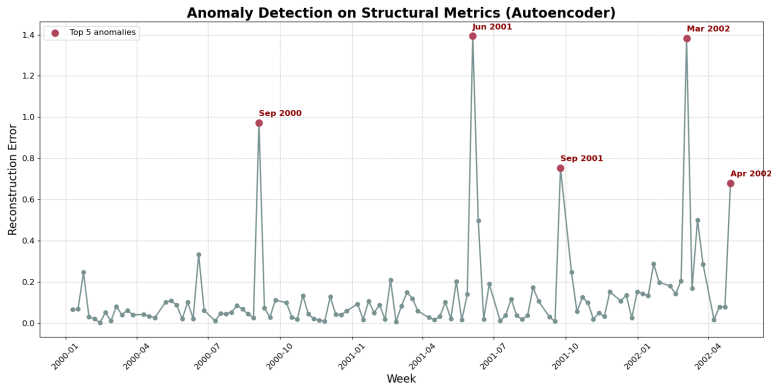


- Significant anomaly peaks are observed in:
 - Sep 2000 First doubts about Enron's (Wall Street Journal article)
 - June 2001 (rising internal conflicts)
 - October 2001 (financial restatements and crisis)
 - February–April 2002 (organizational collapse after bankruptcy)

Autoencoder Applied to Structural Metrics

- Structural metrics for each week were standardized (zero mean, unit variance).
- A standard Autoencoder (MLP) was trained to reconstruct the metrics.
- Anomaly score: Reconstruction error (MSE) for each week.
- Weeks with high reconstruction error are flagged as anomalies.

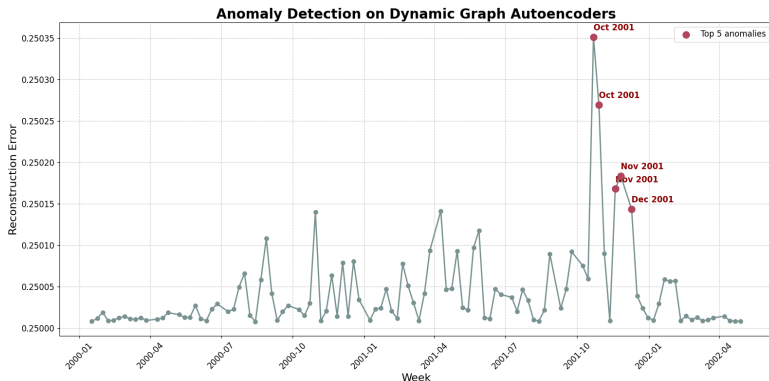
Results: Autoencoder on Structural Metrics



- Major anomaly peaks observed in:
 - Sep 2000 First doubts appear in a Wall Street Journal article
 - June 2001 (emerging internal conflicts)
 - September 2001 (crisis escalation)
 - March–April 2002 (post-bankruptcy structural collapse)

- Architecture consists of:
 - GCN encoder - creates embeddings for node features and edges
 - GRU temporal unit - deals with temporal evolution of graphs
 - decoder - constructs the graph adjacency matrix from embeddings
- The model predicts the next graph in the time sequence
- The predicted graph is compared to the next graph from the dataset
- Anomaly score is calculated based on the MSE loss function

Results: Dynamical Graph Autoencoder



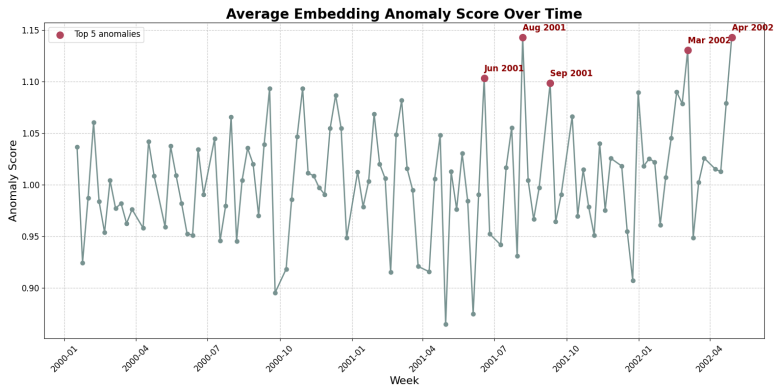
Significant peaks detected around October, November and December 2001.

Node Embedding evolution over time

Idea: If the interactions don't change drastically the embeddings of consecutive weeks should be similar. This method works as follow:

- For each weekly graph, compute an embedding for each node (employee) using a GCN.
- Compute the cosine distance between a node's embeddings at week t and $t + 1$.
- To assign a score to the week the mean of the cosine distance's values of the nodes are computed.
- A large shift indicates a significant behavioral change and so a potential anomaly.

Results: Node Embedding evolution over time

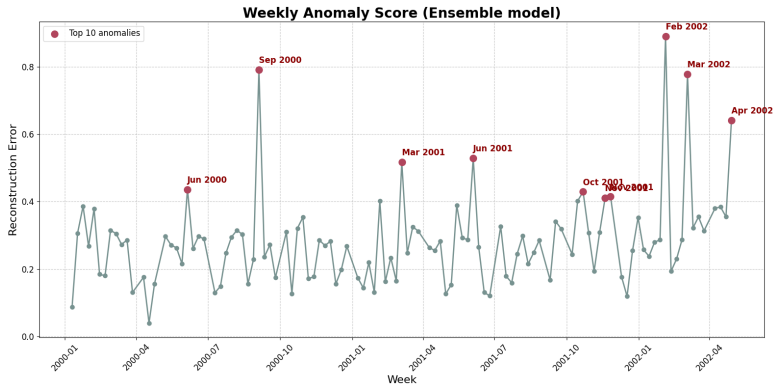


Significant peaks are detected around key dates for the company.

We combined the results of the different models into an ensemble model for a more complete detection.

- Each model generates an anomaly score for each week.
- To ensure that each anomaly score is comparable across different models, the scores are transformed into a consistent scale.
- Once all the anomaly scores are normalized, we average the scores for each week. This is where the ensemble approach comes into play.

Results: Ensemble model



We can observe that the results are spread over a much broader timeframe. This highlights the advantage of combining different approaches.

Section 2

Second Approach: Structure and Content

Methods Applied:

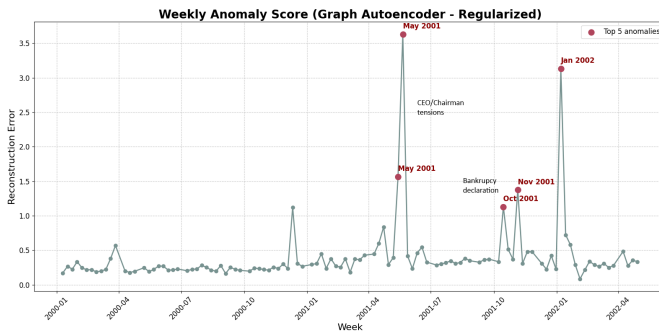
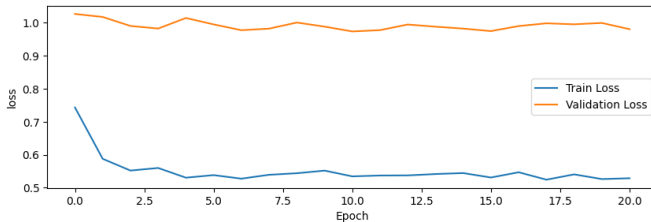
- **Structural Autoencoder I**
- **Structural Autoencoder II**
- **BERT**

The Method: 4 models in total.

- Single instances.
- Train autoencoder (train/validation split).
- Compute reconstruction error.
- Plot the results.

Structural Autoencoder I

Results

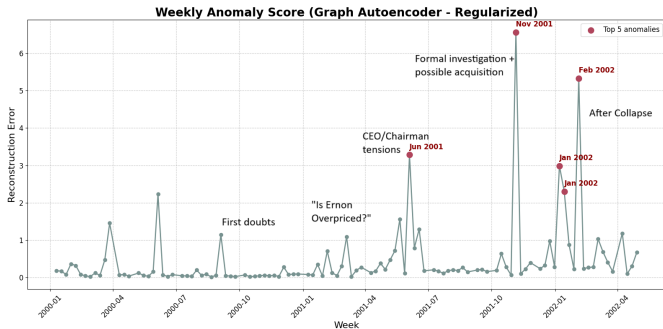
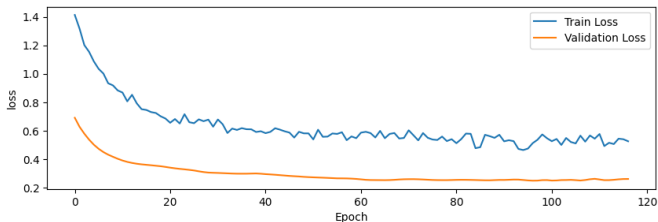


The Method: 2 models in total.

- Weekly grouping.
- Train autoencoder without critical dates.
- Compute reconstruction error.
- Plot the results.

Structural Autoencoder II

Weekly grouping - Results: *Hyper-tuned model.*



--- Emails for week: 2002-02-04 ---

--- Email ID: N/A ---

Please see attached.

Thank you,
Linda and Beverly

Text Inspection

- Noisy Content,
04/02/2002, one
email
- Correlated
Content,
05/11/2001

--- Emails for week: 2001-11-05 ---

--- Email ID: N/A ---

Hi Kam,

This is punit ...i'm working tonite...as i hv been for the past 2 nites. The thing with checking new deals is that i finis

By which time it would have been already too late. Also i have had trades, as you know, which were not mine.. on

I hence request you to undo the penalty for this time as i am setting up all that is needed to be done to trade from

Thanking you.

Regards,

Punit.

Analyzing Email Content with BERT

- We use **BERT (Bidirectional Encoder Representations from Transformers)** to convert each email into a dense vector (embedding).
- Each vector captures the semantic content of the email, its meaning, tone, and structure.
- The model used is `all-MiniLM-L6-v2` (384-dimensional vectors)
- These embeddings allow us to apply standard anomaly detection methods to the content of emails.

Goal: Detect unusual messages or shifts in communication by analyzing their BERT embeddings.

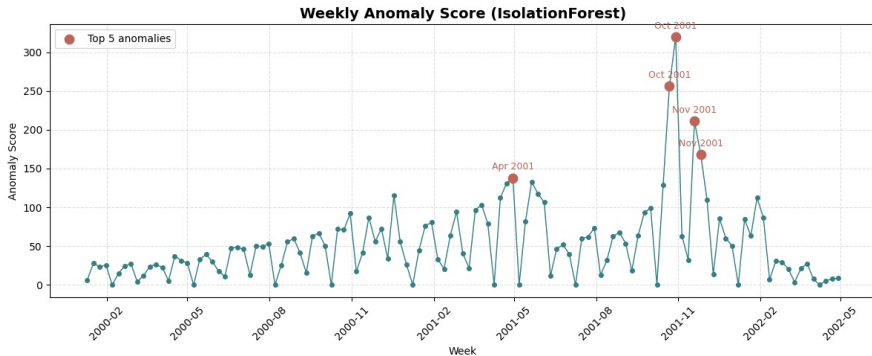
Isolation Forest for Anomaly Detection

- We apply an **Isolation Forest**, an unsupervised model that identifies anomalies by isolating points in the feature space.
- It works by randomly selecting a feature and splitting values: anomalies are easier to isolate (fewer splits).
- Each email's BERT embedding is input into the model.
- Anomaly score is derived from how early the email is isolated in the trees.

$$\text{AnomalyScore}(x) = 2^{-\frac{E(h(x))}{c(n)}}$$

where $E(h(x))$ is the average path length and $c(n)$ is a normalization factor.

Isolation Forest Results



Daily Semantic Shift Detection

- Instead of scoring individual emails, we also detect semantic shifts across days.
- For each day, compute the average BERT embedding of all emails sent.
- Then compute the Euclidean distance to the previous day's average.

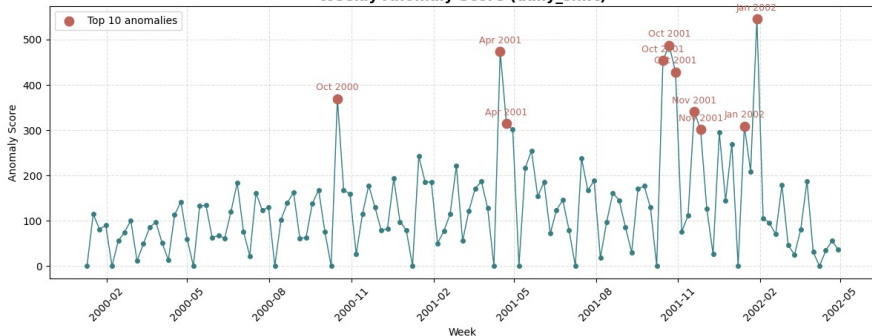
$$\text{Shift}(d) = \|\mu_d - \mu_{d-1}\|_2$$

- Large shifts indicate changes in tone, topic, or behavior.

Note: Days with sudden change in content are flagged, not individual emails.

Daily Shift Results

Weekly Anomaly Score (daily_shift)



Conclusion

- Structure-based and content-based approaches
- Graph models: GAE, metrics, autoencoders
- Content models: BERT embeddings + Isolation Forest
- Daily shift: detects group-level changes
- Peaks match scandal timeline
- Different models find different moments
- Noisy data

Thank you for the...

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax} \left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}} \right) \mathbf{V} \in \mathbb{R}^{m \times d_v}$$

Section 3

Appendix

Text inspection - Structural metrics, content approach.

-----> NEXT MAIL <-----

Anomaly Score: 366.23948025271756

Date: 2001-11-12 12:30:11

Email Body:

I received another call from Robert Minkoff on Friday, 11/09. His company is interested in purchasing EPMI's claim in the PCA bankruptcy.

Bonnie J. White

Enron Litigation Unit

1400 Smith Street, Suite 4824

Houston, TX 77002

Ph: 713-853-7244

Fax: 713-853-6576

bonnie.white@enron.com

Structure and Content Approach

Content Pre-Processing: from 715398 to 105332 instances.

- Duplicates Removal

	sender	receiver	date	subject	body
0	jeff.youngflesh@enron.com	anthony.gillmore@enron.com	Mon, 27 Nov 2000 01:49:00 -0800 (PST)	Invitation: EBS/GSS Meeting w/Bristol Babcock ...	Conference Room TBD. \n\nThis meeting will be...
1	jeff.youngflesh@enron.com	colleen.koenig@enron.com	Mon, 27 Nov 2000 01:49:00 -0800 (PST)	Invitation: EBS/GSS Meeting w/Bristol Babcock ...	Conference Room TBD. \n\nThis meeting will be...
2	jeff.youngflesh@enron.com	jennifer.stewart@enron.com	Mon, 27 Nov 2000 01:49:00 -0800 (PST)	Invitation: EBS/GSS Meeting w/Bristol Babcock ...	Conference Room TBD. \n\nThis meeting will be...

- Useless text

I had Devon put together a resume for you to get an idea of his experience to date. He's not our typical MBA Associate, but I feel he has great potential as a junior trader. I'll call you later to discuss. Per

----- Forwarded by Per Sekse/NY/ECT on 11/10/2000 04:07 PM

Enron Capital & Trade Resources Corp.

From: Devin_Burnett@PECHINEY.COM

11/10/2000 02:17

PM

To: psekse@ect.enron.com

cc:

Subject:

Structure and Content Approach

Structure Pre-Processing: from 715398 to 114614 instances.

- Duplicates Removal
- Feature Extraction

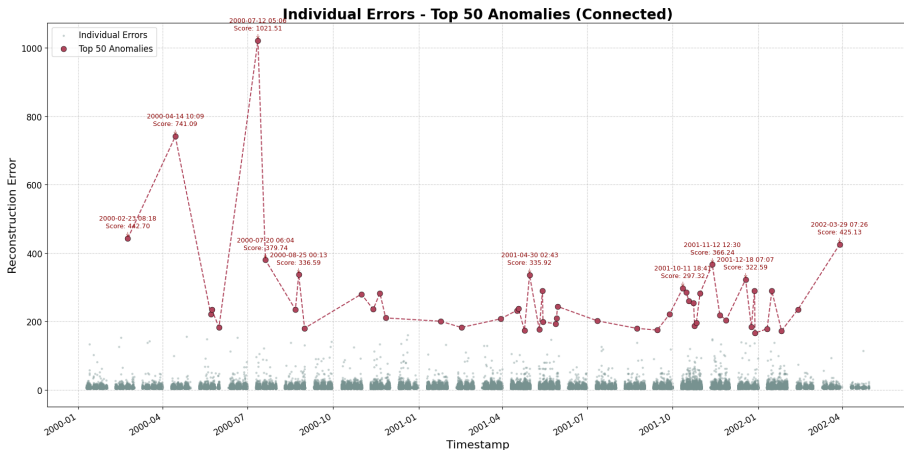
```
df_features.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 114614 entries, 0 to 114613
Data columns (total 26 columns):
#   Column                Non-Null Count  Dtype
---  -
0   char_count             114614 non-null  int64
1   word_count             114614 non-null  int64
2   sentence_count         114614 non-null  int64
3   avg_sentence_length     114614 non-null  float64
4   max_sentence_length     114614 non-null  int64
5   min_sentence_length     114614 non-null  int64
6   paragraph_count        114614 non-null  int64
7   num_exclamations       114614 non-null  int64
8   num_questions          114614 non-null  int64
9   num_commas             114614 non-null  int64
10  num_periods            114614 non-null  int64
11  num_colons              114614 non-null  int64
12  capital_ratio           114614 non-null  float64
13  special_char_ratio      114614 non-null  float64
14  line_count              114614 non-null  int64
15  avg_line_length         114614 non-null  float64
16  std_line_length         114614 non-null  float64
17  greeting_dear           114614 non-null  int64
18  signoff_regards         114614 non-null  int64
19  has_forward             114614 non-null  int64
20  has_original            114614 non-null  int64
21  has_disclaimers         114614 non-null  int64
22  quotations              114614 non-null  int64
23  automatically_generated 114614 non-null  int64
24  week                   114614 non-null  object
25  parsed_date             114614 non-null  object
```

Structural Metrics

- Single instance
- Compute standardized z-scores for each feature.
- Define the anomaly score as the sum of absolute z-scores.

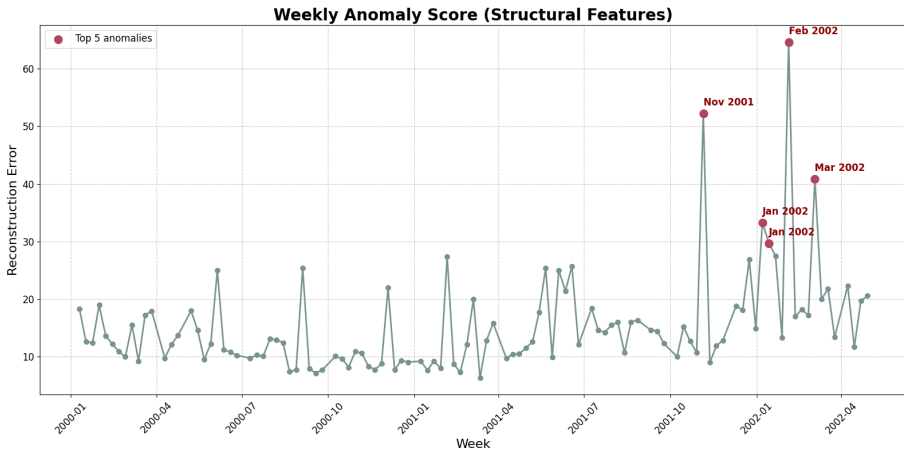
Single instance



Structural Metrics

- Weekly grouping
- Compute standardized z-scores for each feature.
- Define the anomaly score as the sum of absolute z-scores.

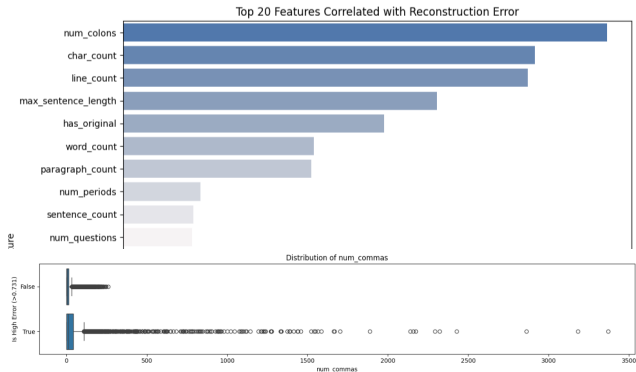
Weekly grouping



Structural Autoencoder I

Explainability

- Feature differences
- Direct correlation
- Feature distribution comparison
- Peak averages
- Common Features



Features in common:

{'char_count', 'line_count', 'num_colons', 'word_count'}

Text Inspection

- Noisy Content,
21/05/2001
- Correlated
Content,
05/11/2001

```
<OMNICALendarEntry>
  <OMNICALendarEntryID>1040</OMNICALendarEntryID>
  <OMNICreatedDate>10/11/2000 12:50:40 PM</OMNICreatedDate>
  <OMNIForm>Appointment</OMNIForm>
  <OMNIExcludeFromView>D</OMNIExcludeFromView>
  <OMNIS$PublicAccess>1</OMNIS$PublicAccess>
  <OMNIChair>CN=3DJohn M Forney/OU=3DHOU/O=3DECT</OMNIChair>
  <OMNIOrgTable>C0</OMNIOrgTable>
  <OMNIS$BusyName>CN=3DJohn M Forney/OU=3DHOU/O=3DECT</OMNIS$BusyName>
  <OMNIS$Busy>1</OMNIS$Busy>
  <OMNIS$BusyPriority>1</OMNIS$BusyPriority>
  <OMNIPrincipal>CN=3DJohn M Forney/OU=3DHOU/O=3DECT</OMNIPrincipal>
  <OMNIAppointmentType>0</OMNIAppointmentType>
  <OMNI_ViewIcon>160</OMNI_ViewIcon>
  <OMNICALendarDateTime>10/16/2000 01:00:00 PM</OMNICALendarDateTime>
  <OMNIStartDate>10/16/2000 01:00:00 PM</OMNIStartDate>
  <OMNIStartDateTime>10/16/2000 01:00:00 PM</OMNIStartDateTime>
  <OMNIOriginalDateTime>10/16/2000 01:00:00 PM</OMNIOriginalDateTime>
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  <OMNIS$NoPurge>10/16/2000 02:00:00 PM</OMNIS$NoPurge>
  <OMNITimeRange>10/16/2000 01:00:00 PM-10/16/2000 02:00:00 PM</OMNITimeRange>
```

A full list of news articles will be distributed on Monday, but here is some of the initial coverage.

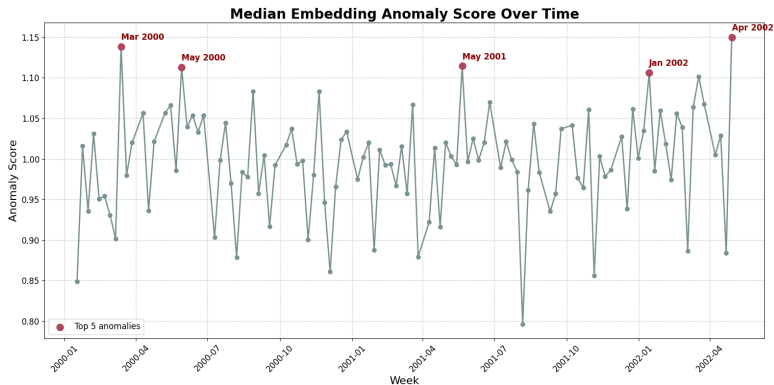
Business/Financial Desk; Section A
Rival to Buy Enron, Top Energy Trader, After Financial Fall
By ALEX BERENSON and ANDREW ROSS SORKIN

11/10/2001
The New York Times=20
Page 1, Column 2
c. 2001 New York Times Company=20

With its stock plunging and its finances in doubt, the world's largest energy trader, the Enron Corporation, agreed to be acquired yesterday by the rival Dynegy Inc. for about \$9 billion in stock and the assumption of \$13 billion in debt.=20

Nodel Embedding evolution over time - median

We tried to use the **median** (less sensitive to outliers) instead of the average, but the results were still noisy.



Further Approaches

- Combining both structural and content information in the embeddings.
- Learning better weights for the ensemble model.
- Exploring additional measures of nodes (betweenness, closeness,...).
- Improve pre-processing on structural features.

Examples of Noise

📌 Anomalous day: 2000-04-30 with 4 emails

--- Email 5562 ---

From: karen.denne@enron.com

Content: BusinessWeek is writing a story for its eBiz insert in mid-May about how Enron is using the Internet and e-commerce to again transform its businesses. Wendy Zellner, who works out of the magazine's Dallas bureau, will be in Houston on Tuesday and Wednesday to conduct interviews.

The following are the interview topics and schedule:

Tuesday, May 2

10 a.m. – Shareholder meeting

3:30 p.m. – Louise Kitchen

How was EnronOnline conceived and created?

What did you and your group have to do to get [...]

--- Email 5563 ---

From: carrie.robert@enron.com

Content: Attached is the final agenda topics and attendees for the Gaz de France visit to Enron Monday, May 1, 2000. Below a few other pertinent details.

First

As you will see in the agenda, a GDF representative will be giving a brief presentation at the very start of the Briefing This person will highlight:

Where GDF stands now in the market place

What their rates are and how they stack up against the competition

What they hope to get out of Monday's meeting with Enron

NOTE – Past briefings [...]

Figure: Example of Anomaly day